HR Analytics Pipeline

Employee Attrition & Workforce Analysis Report

MySQL to Python: End-to-End HR Data Analytics Project

Data Analytics Internship Portfolio Project

Prepared: July 2026

Note: This analysis is based on a sample HR dataset of 915 employee records. See Data Disclaimer (Section 14) for full context on scope and limitations.

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Section 1: Introduction & Objective

Introduction

This report presents a complete Human Resources analytics pipeline built end to end, from a raw MySQL database through data cleaning, verification, exploratory data analysis, and finally into charts and written business insights. The underlying dataset covers 915 employee records spread across eight departments, drawn from a relational schema of eight linked tables covering employees, jobs, departments, locations, job history, performance reviews, attendance, and attrition.

The goal of this project was to simulate a realistic company HR analytics workflow: pull messy operational data out of a database, clean it defensibly, verify it, and turn it into decisions a people team could actually act on.

Objective

- Understand overall attrition and identify which departments and employee groups are most at risk.
- Examine whether compensation, tenure, and overtime patterns differ between active and exited employees.
- Test whether job satisfaction is actually linked to performance ratings, rather than assuming it is.
- Identify the highest-paying roles and the departments carrying the most headcount.
- Translate these patterns into concrete retention and workforce planning recommendations.

Tools & Environment

Tool / Library	Purpose
Python 3	Core programming language for the pipeline
Pandas	Data cleaning, transformation, and aggregation
NumPy	Numerical calculations behind the written insights
Matplotlib & Seaborn	Chart generation
SQLAlchemy & PyMySQL	Database connection and queries
MySQL	Source relational database (8 tables)

Section 2: Dataset Overview

The data originates from a purpose-built MySQL schema representing a mid-sized company's HR system. It is a relational database of eight tables connected by employee, job, and department identifiers, designed to mirror how a real HR system separates core employee records from performance, attendance, and exit data.

Database Schema

Table	Rows (Raw)	Rows (Cleaned)	Purpose
locations	12	12	City, state, and country reference data
departments	10	8	Department names, linked to a location. Two duplicate entries merged
jobs	30	30	Job titles with defined min/max salary bands
employees	915	915	Core employee records: demographics, salary, role, manager
job_history	1,307	1,307	Historical record of role and department changes per employee
performance_reviews	1,848	1,848	Performance rating, job satisfaction, work-life balance scores
attendance	8,400	8,400	Daily attendance status and overtime hours logged
attrition	146	146	Exit date and reason for employees who have left

The employees table is the anchor: every one of the 915 employee records was retained through cleaning. The departments table dropped from 10 to 8 rows because two departments ("HR" and "Ops") had been entered twice under different IDs and were merged into their full names, "Human Resources" and "Operations", during cleaning.

Master Table

After cleaning, all eight tables were verified and joined into a single master_table.csv of 915 employee rows by 26 columns, combining core demographics, current job and department, latest performance review, latest attendance record, and attrition status into one analysis-ready table. This master table is the source for every chart and insight in this report.

Column Reference (Key Fields)

Column	Type	Description
employee_id	Integer	Unique employee identifier
gender	Text	Male / Female
hire_date	Date	Date the employee joined

Column	Type	Description
salary	Numeric	Annual salary
education_level	Text	High School, Diploma, Bachelor's, Master's, PhD
business_travel	Text	Non-Travel, Travel_Rarely, Travel_Frequently
department_name	Text	One of 8 departments
job_title	Text	One of 30 job titles, each with a defined salary band
performance_rating	Numeric (1-5)	Latest performance review score
job_satisfaction	Numeric (1-5)	Latest job satisfaction score
work_life_balance	Numeric (1-5)	Latest work-life balance score
overtime_hours	Numeric	Overtime hours logged on the latest attendance record
is_active	Boolean	True if still employed, False if exited
exit_reason	Text	Reason for leaving, where applicable

Section 3: Data Pipeline Architecture

The pipeline was built as five sequential Python stages, each depending on the output of the one before it. This mirrors a production analytics workflow where raw data has to be cleaned and verified before it can be trusted for reporting.

Pipeline Flow

Stage	Script	What It Does
1. Fetch	fetch.py	Builds the database connection from environment credentials
2. Clean	cleaning.py	Reads each raw table, applies cleaning rules, saves *_cleaned.csv
3. Explore	EDA.py	Generates exploratory stats and per-table diagnostic plots
4. Verify	verification.py	Re-checks cleaned data, inserts into a clean database, builds the master table and subset CSVs
5. Report	visualization.py / insights.py	Produces the 5 final charts and the 8 written HR insights

Each script only processes one table at a time, called manually in sequence, so that a failure in one table (for example, a verification check on employees) does not silently corrupt the rest of the pipeline. finalize_pipeline() only builds the master table once every one of the eight tables has passed verification.

Section 4: Data Cleaning: Steps & Decisions

Every cleaning decision below was made for a specific, documented reason rather than applied as a blanket rule. The two guiding principles were: fix data that is clearly wrong instead of deleting it where possible, and never let a formatting bug silently duplicate or lose a record.

Cleaning Steps Applied

Step	Action	Reason
1	Standardize text casing (city, state, department, job title)	Raw entries mixed cases such as "MUMBAI", "kolkata", "it manager"
2	Fix acronyms broken by title-casing	Title-casing turned "IT" into "It" and "HR" into "Hr"; corrected back
3	Merge duplicate departments (HR/Ops)	Same real department entered under two different IDs; remapped and merged into the full name
4	Standardize gender labels	Raw data mixed "M", "F", "male", "female"; mapped to Male/Female
5	Fill missing email with a flag	Cannot guess a missing email; labeled MISSING rather than left blank
6	Fill missing salary by job role median	Missing salaries filled using the median salary for the same job_id, falling back to the overall median if needed
7	Correct negative distance_from_home_km	EDA found negative values (minimum of -52); corrected using absolute value rather than dropped
8	Correct salaries outside job salary band	Salaries below a job's min_salary were raised to the minimum; salaries above max_salary were capped, with each correction counted rather than silently clipped
9	Clip negative overtime_hours to zero	Overtime hours cannot be negative; any negative values were clipped to 0
10	Fill missing attrition reason	Missing exit reasons labeled "Not Specified" instead of left blank
11	Downcast numeric dtypes, convert low-cardinality text to category	Reduces memory footprint without losing precision
12	Drop duplicate rows per table	Applied after all other cleaning, confirmed 0 true duplicates remained

Known Data Limitations After Cleaning

Cleaning corrected structural errors, but a few limitations remain by design rather than by oversight, and are relevant to how confidently the results below should be read:

- manager_id is missing for 94 of 915 employees. This is expected: top-level employees have no manager, so these were intentionally left blank rather than imputed.

- overtime_hours has no attendance record for 515 of 915 employees in their latest snapshot, since not every employee has a matching latest-day attendance row. Overtime comparisons in this report use only the 400 employees with a valid value.
- performance_rating and work_life_balance are missing for 83 and 61 employees respectively where no review had yet been logged; these rows are excluded from calculations involving those specific fields rather than imputed.
- The attendance status field still contains some short-form entries ("P", "A", "Wfh") alongside the standardized long-form labels ("Present", "Absent", "Work From Home"), reflecting inconsistent entry at source that full standardization did not fully resolve.

Section 5: Exploratory Data Analysis

Key Metrics Summary

Metric	Value
Total employees analyzed	915
Active employees	769 (84.0%)
Exited employees	146 (16.0%)
Average salary	61,266
Median salary	61,079
Average tenure (all employees)	8.13 years
Average tenure (exited employees only)	6.18 years

Salary Distribution

Statistic	Value
Min	27,000
25th percentile	52,442
Median	61,079
Mean	61,266
75th percentile	70,000
Max	90,000
Std deviation	13,074

Department Headcount

Department	Employees	Share
Sales	126	13.8%
Research & Development	125	13.7%
Human Resources	121	13.2%
IT	115	12.6%
Finance	112	12.2%
Customer Support	107	11.7%
Operations	106	11.6%
Marketing	103	11.3%

Workforce Profile

Category	Breakdown
Gender	Female 491 (53.7%) | Male 424 (46.3%)
Education level	Master's 198 | Diploma 194 | High School 178 | PhD 175 | Bachelor's 170
Business travel	Non-Travel 307 | Travel_Rarely 305 | Travel_Frequently 303
Location (top cities)	Chennai 231 | Lucknow 224 | Bangalore 126 | Jaipur 115 | Mumbai 112 | Noida 107

Section 6: Visualization 1: Attrition Rate by Department

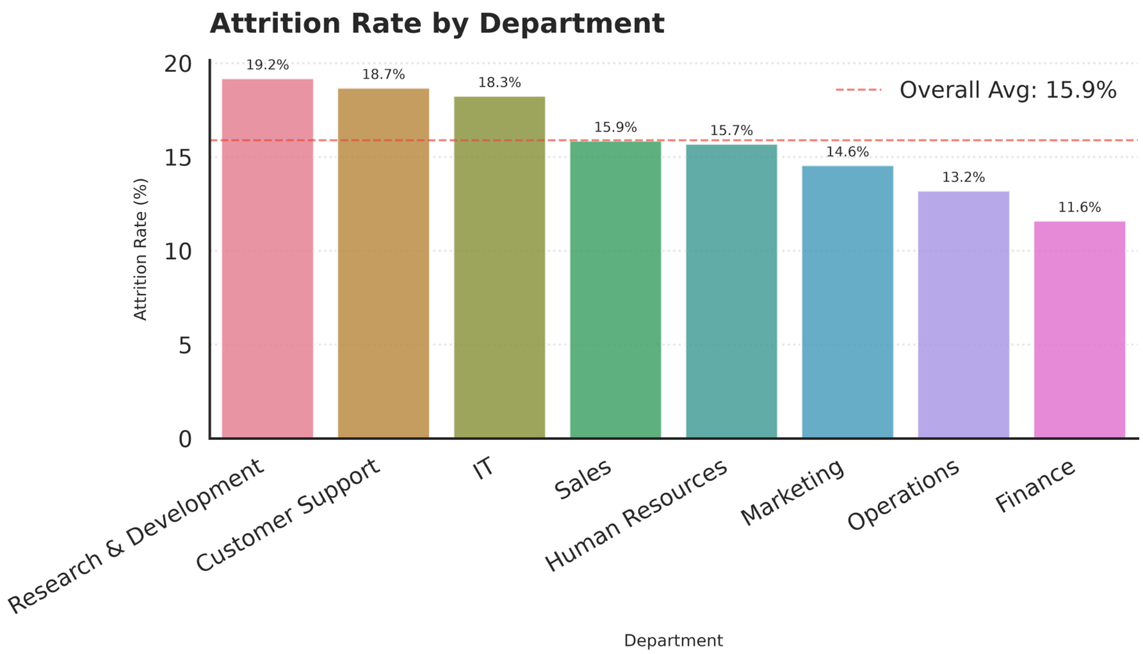

Fig 1. Attrition rate by department, with the overall company average shown as a dashed reference line.

What does this chart mean?

Each bar shows the percentage of employees in that department who have exited the company, calculated as exited employees divided by total employees in that department. The dashed red line marks the average attrition rate across all eight departments, so any bar above the line is a department losing people faster than the company average.

Key Takeaways

- **Research & Development runs hottest:** R&D has the highest attrition rate at 19.20%, followed closely by Customer Support (18.69%) and IT (18.26%).
- **Finance and Operations are the most stable:** Finance has the lowest attrition rate at 11.61%, with Operations close behind at 13.21%, both well under the company average.
- **The gap is not small:** The difference between the highest and lowest department attrition rates is roughly 7.6 percentage points, which is a meaningful spread for a workforce this size.

Section 7: Visualization 2: Salary Dispersion, Active vs Exited Employees

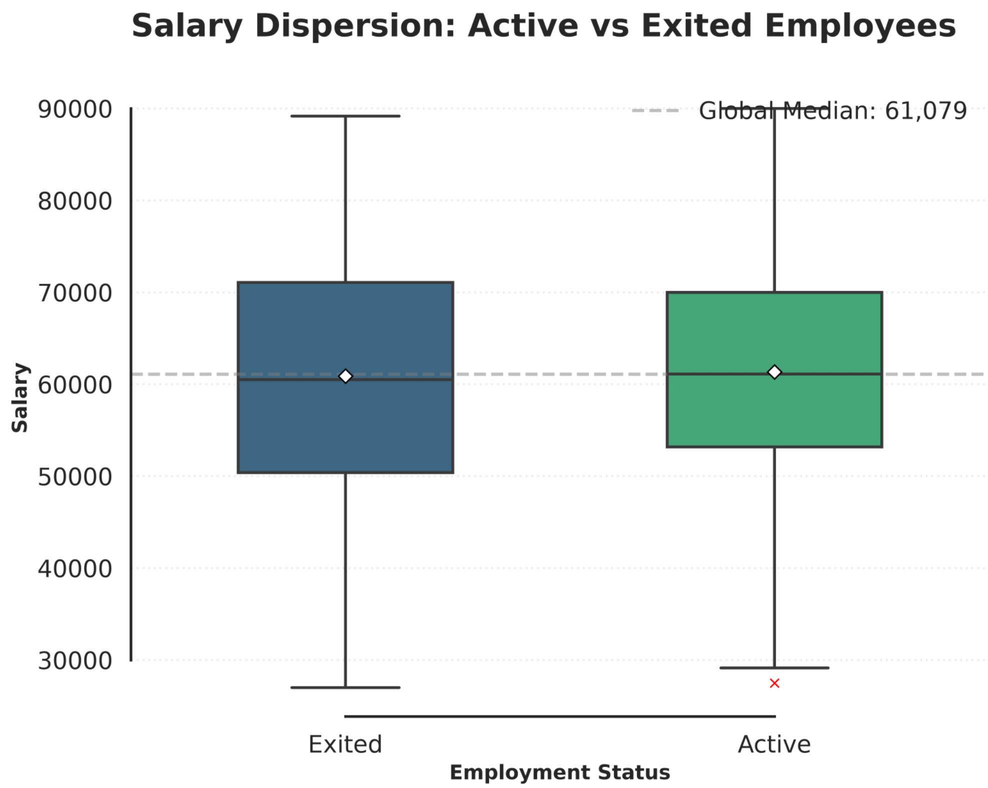

Fig 2. Box plot comparing salary distributions between active and exited employees. Diamond markers show the mean; the dashed line shows the global median.

What does this chart mean?

This box plot lays the full salary distribution of active employees next to exited employees. The box shows the middle 50% of salaries, the line inside the box is the median, and the diamond marker is the mean. If exited employees were clustered at the bottom of the pay scale, we would expect their box to sit visibly lower than the active employees' box.

Key Takeaways

- **Pay is not the main driver of exits:** Active employees earn 452 more on average than exited employees (mean of 61,338 vs 60,886), a gap of well under 1%. That is not a meaningful salary difference.
- **Medians tell the same story:** Median salary for active employees is 61,116 versus 60,506 for exited employees, again a very small gap relative to the overall salary range of 27,000 to 90,000.
- **Implication:** Because the salary distributions of active and exited employees overlap almost completely, compensation alone is unlikely to explain why people are leaving. The cause is more likely to sit in role, department, or workload factors covered elsewhere in this report.

Section 8: Visualization 3: Employee Tenure Distribution

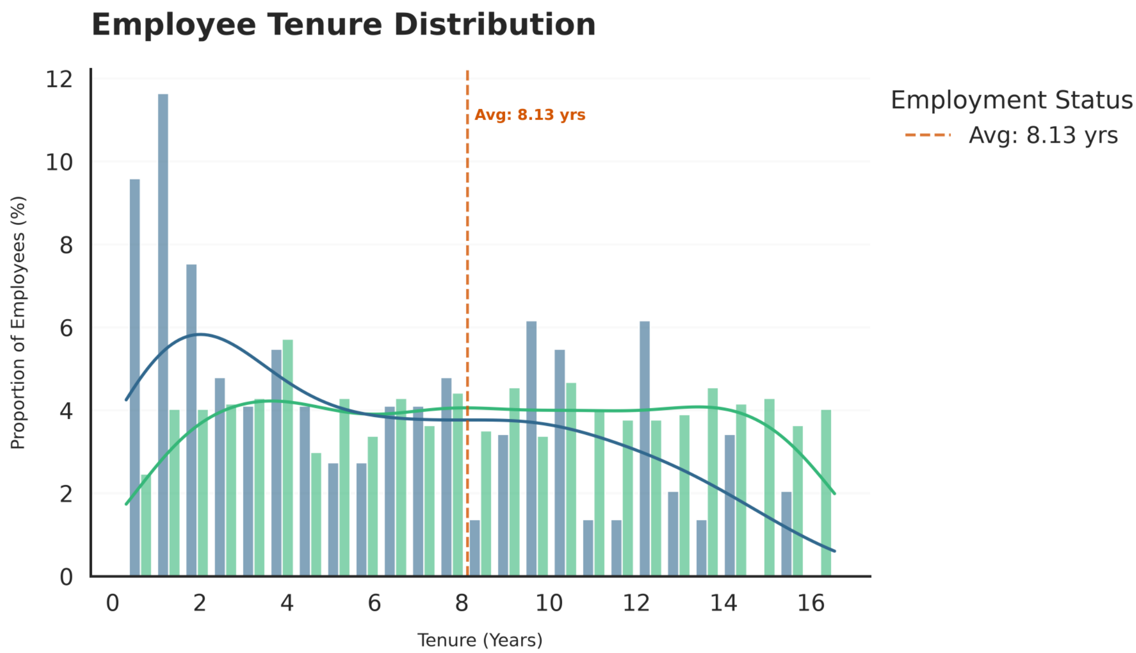

Fig 3. Distribution of employee tenure in years, split by employment status, with the average tenure marked.

What does this chart mean?

This chart shows how long employees stay, comparing the shape of the tenure distribution for active employees against exited employees. The dashed line marks the average tenure across the full workforce.

Key Takeaways

- **Employees who leave, leave earlier:** Employees who exited stayed for 6.18 years on average, well short of the company-wide average tenure of 8.13 years.
- **Attrition risk is front-loaded:** The gap between overall and exited-only average tenure suggests the exit risk is concentrated in the earlier and mid stages of an employee's time at the company rather than spread evenly across all tenure lengths.
- **Retention window:** The first six to eight years appear to be the critical retention window; employees who make it well past that point are more likely to represent the long-tenured, currently active population.

Section 9: Visualization 4: Feature Correlation Heatmap

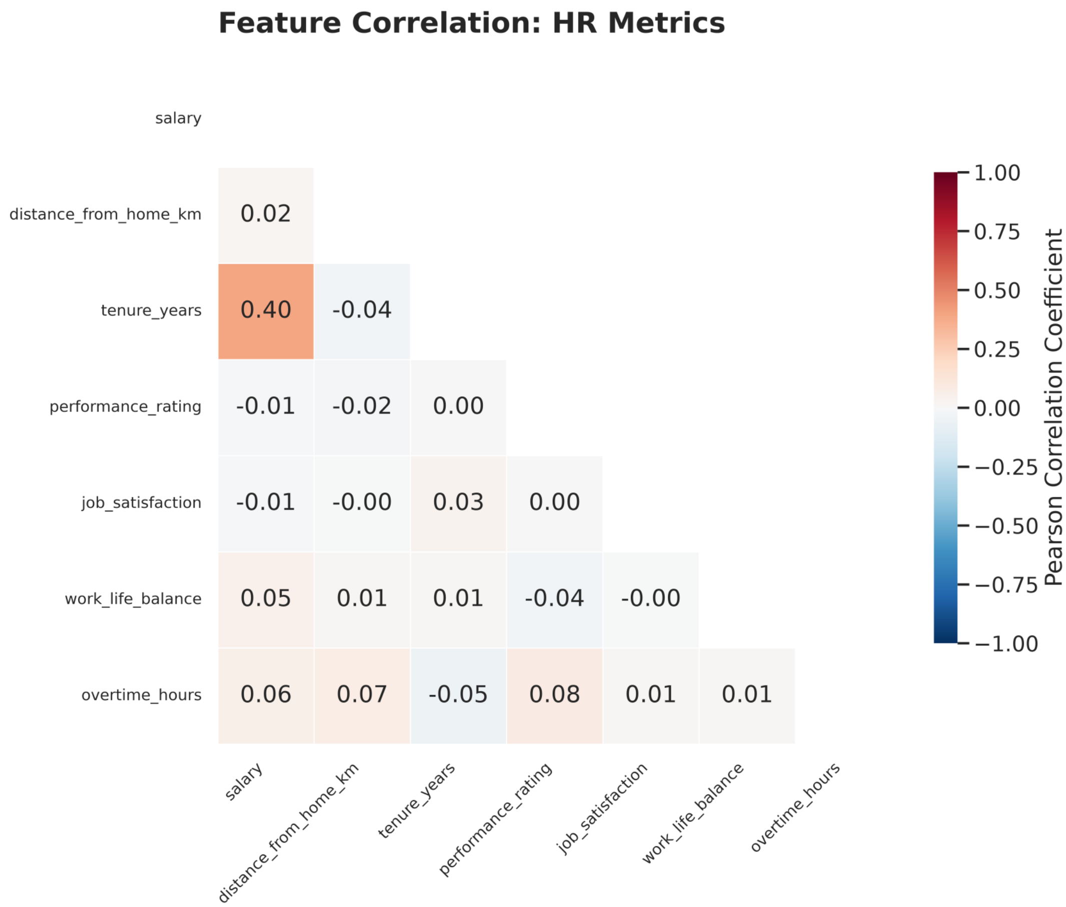

Fig 4. Pearson correlation coefficients across salary, tenure, performance, satisfaction, work-life balance, distance from home, and overtime.

What does this chart mean?

This heatmap shows how strongly every pair of numeric HR metrics moves together, using Pearson correlation. Values close to 1 or -1 indicate a strong relationship; values close to 0 indicate little to no linear relationship. Only the values below the diagonal are shown to avoid repeating each pair twice.

Key Takeaways

- **Tenure and salary are genuinely linked:** Salary and tenure show a correlation of 0.40, the strongest relationship on the whole board, which makes sense given standard pay progression over time.
- **Happiness does not predict performance:** Job satisfaction and performance rating correlate at essentially 0.00, confirming that in this dataset, how satisfied an employee reports being has no measurable relationship with how they are rated.
- **Everything else is close to flat:** Overtime hours, distance from home, and work-life balance all show weak correlations (under 0.10 in magnitude) with salary and performance, meaning none of them move in a strong linear pattern with compensation or rating outcomes in this dataset.

Section 10: Visualization 5: Headcount by Department & Gender

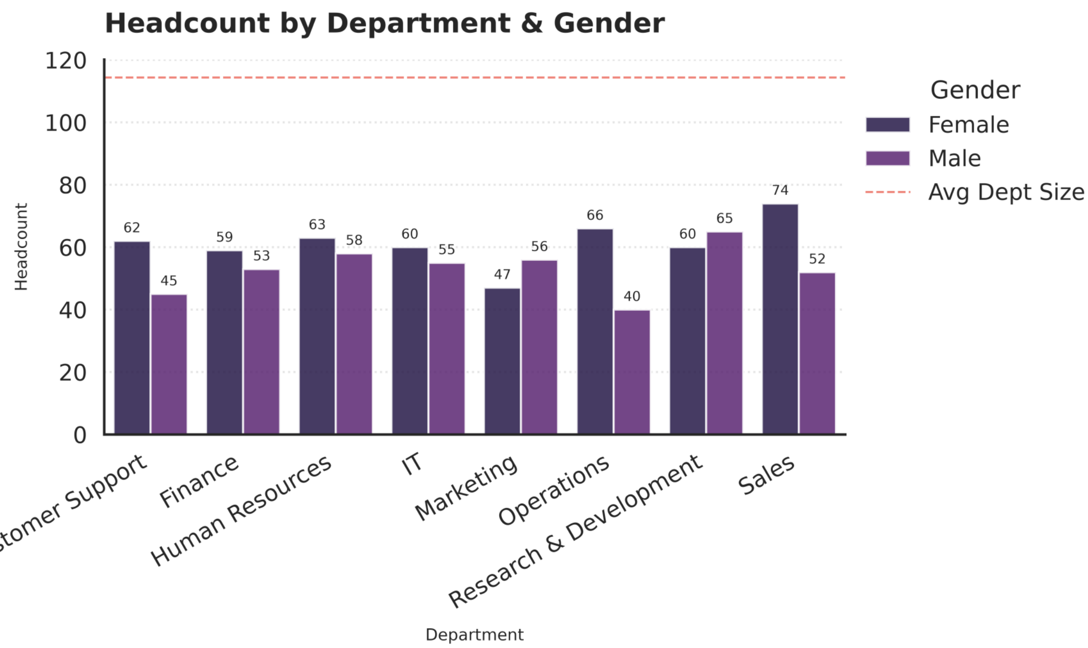

Fig 5. Total headcount per department, split by gender, with the average department size shown as a dashed reference line.

What does this chart mean?

Each pair of bars breaks down department headcount by gender, letting us see not just which departments are largest but whether any department skews heavily toward one gender.

Key Takeaways

- **Departments are close to evenly sized:** All eight departments fall within a narrow band of roughly 103 to 126 employees, so no single department dominates total headcount.
- **Gender split is broadly balanced company-wide:** The overall workforce is 53.7% female and 46.3% male, and this balance holds reasonably consistently across most departments rather than being concentrated in one or two.
- **Useful as a baseline:** Because department sizes are close to even, attrition rate comparisons between departments earlier in this report are not distorted by one department being disproportionately large or small.

Section 11: HR Insights: Eight Key Questions Answered

The eight questions below were run directly against the 915-row master dataset using NumPy, and are reproduced here with the exact figures the pipeline produced.

Q1. What is the overall attrition rate?

15.96% of employees have left the company (146 out of 915).

Q2. Which department has the highest attrition rate?

Research & Development has the highest attrition rate at 19.20%.

Q3. Is there a salary gap between active and exited employees?

Active employees earn 452 more on average than exited employees. Active: mean 61,338, median 61,116. Exited: mean 60,886, median 60,506. The gap is small relative to the overall salary spread.

Q4. What is the average employee tenure?

On average, employees stay 8.13 years. Employees who left stayed 6.18 years on average before leaving.

Q5. Is attrition different between genders?

Male employees leave more often, at 16.75%, compared with 15.27% for female employees. The gap is present but modest.

Q6. Are happier employees also better performers?

The correlation between job satisfaction and performance rating is 0.00. This means there is almost no measurable link between the two in this dataset.

Q7. Which job title earns the most on average?

IT Manager has the highest average salary at 78,599, followed closely by Software Engineer (78,054) and System Admin (77,652).

Q8. Do employees who work more overtime leave more often?

Active employees log 1.08 average overtime hours versus 1.15 for exited employees. Employees who left worked more overtime on average, which suggests overwork may be linked to attrition, though the difference is modest and based on the 400 employees with a recorded overtime value.

Section 12: Real-World Problems & Solutions

Based on the patterns above, three concrete workforce problems stand out, each with a data-backed direction for action.

Problem 1: R&D, Customer Support, and IT Are Bleeding Talent Faster Than the Rest of the Company

- **The Data:** Research & Development (19.20%), Customer Support (18.69%), and IT (18.26%) all sit well above the company-wide attrition rate of 15.96%, while Finance (11.61%) and Operations (13.21%) are comfortably below it.
- **The Solution:** Since the salary gap between active and exited employees is negligible (452 on average), pay is unlikely to be the fix here. These three departments warrant targeted exit interviews and workload or role-clarity reviews rather than a blanket compensation response.

Problem 2: Attrition Risk Is Concentrated in the First Six to Eight Years

- **The Data:** Employees who leave do so after an average of 6.18 years, well short of the company-wide average tenure of 8.13 years.
- **The Solution:** Retention efforts, career development check-ins, and internal mobility programs are likely to have the most impact if concentrated on employees in their third through seventh year, rather than applied uniformly across all tenure bands.

Problem 3: Overtime Load Tracks With Who Leaves

- **The Data:** Employees who exited logged 1.15 average overtime hours on their latest attendance record versus 1.08 for active employees. The correlation matrix also shows overtime has a small positive relationship with performance rating (0.08), meaning higher performers may be the ones absorbing more overtime.
- **The Solution:** Combine overtime tracking with performance and attrition flags so managers can spot when a strong performer's overtime load is trending upward, before it becomes a resignation.

Section 13: Strategic Recommendations

- **Prioritize R&D, Customer Support, and IT for retention review:** these three departments run 2 to 3 percentage points above the company attrition average and should be the first target for exit interviews and manager 1:1 reviews.
- **Do not lead with compensation as the retention fix:** the salary gap between active and exited employees is under 1%, so pay adjustments alone are unlikely to move the needle. Investigate workload, role clarity, and management quality instead.
- **Build a tenure-based retention program:** concentrate retention and development conversations on the three-to-seven-year tenure band, since that is where exits are concentrated relative to the company-wide average tenure of 8.13 years.
- **Do not treat job satisfaction scores as a performance proxy:** with a correlation of 0.00 between satisfaction and performance rating, these should be tracked and managed as two separate signals, not used interchangeably in performance conversations.
- **Watch overtime trends for high performers specifically:** since exited employees carried a higher average overtime load, pair overtime tracking with performance data to catch burnout risk in your strongest employees before they leave.

Section 14: Data Disclaimer & Conclusion

Conclusion

This HR analytics pipeline takes 915 employee records through a full cleaning, verification, and analysis workflow, from a raw eight-table MySQL schema to five recruiter-ready charts and eight quantified business insights. The clearest finding is that attrition in this dataset is not primarily a pay problem. Salary differences between active and exited employees are marginal, while department, tenure stage, and overtime load show a much stronger relationship with who leaves.

The project demonstrates a complete analytics workflow: defensible data cleaning with each decision justified by evidence rather than convenience, memory-conscious processing, relational joins across eight tables into a single master dataset, and translation of raw statistics into specific, actionable HR recommendations.

Data Disclaimer & Scope

This report is based on a sample HR dataset of 915 employee records generated for a data analytics portfolio project, not a live production HR system. The dataset was designed to include realistic data quality issues, such as inconsistent casing, duplicate department entries, out-of-range salaries, and missing values, so that the cleaning steps documented in Section 4 could be demonstrated and justified rather than assumed.

A number of fields have partial coverage rather than complete records, most notably overtime_hours (present for 400 of 915 employees) and performance_rating (missing for 83 employees). Every statistic in this report that uses these fields is calculated only from the employees with a valid value for that field, and the relevant sample size is called out where it materially affects interpretation.

Because this is a fixed-size sample dataset rather than a live, continuously updated HR system, the findings here should be read as a demonstration of analytical method and a directional read on the patterns in this specific dataset, not as a substitute for a live HR system's ongoing monitoring.